

Systems of Equations with Matrices

POSSIBLE SOLUTION SETS

CONSISTENT SOLUTION SETS

FINITE SET

$$-5x - 6y + 2z = -19$$

$$-2x + 3y - 3z = -25$$

$$-6x + 6y + 2z = -12$$

(5, 1, 6)

$$\begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 6 \end{bmatrix}$$

INFINITE SET

$$-r - s + 2t = 5$$

$$-2r - 4s + 2t = -6$$

$$2r + 2s - 4t = -10$$

Infinitely many solutions

$$\begin{bmatrix} 1 & 0 & -3 & -13 \\ 0 & 1 & 1 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

INCONSISTENT SOLUTION SETS

NO SOLUTION (THE EMPTY SET)

$$4a + 4b - 6c = -12$$

$$2a + 2b - 3c = -18$$

$$a - 4b - 5c = -4$$

No solution

$$\begin{bmatrix} 1 & 0 & -2.2 & 0 \\ 0 & 1 & 0.7 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$